Module 4 — Assembly: Theoretical Compendium

BruSafe / CAPSID · iGEM 2026 · the letter A in CAPSID

A single coherent derivation of every equation Module 4 rests on, in the order the physics requires, with each parameter traced to a Crossref-verified source and each numerical claim reproduced from running code.

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0. What this document is, and what it is not

Module 4 answers one question: **given a pool of MS2 coat-protein dimers, what fraction closes into well-formed T=3 capsids, and at what dimer concentration is that fraction maximal?**

That question has been attacked at four different levels of description, each of which exists in this project as running code rather than as prose. This compendium is the theory behind all four, assembled as one argument:

Layer	Level of description	State variable	Answers
L1	Deterministic rate equations	Concentration of each partial shell size	Yield vs concentration and time
L2	Classical nucleation theory	Cluster size as a continuous coordinate	*Where* in parameter space assembly is possible at all
L3	Coarse-grained Monte Carlo	Positions and orientations of 90 subunits	The geometry L1 must assume
L4	Stochastic co-assembly with cargo	Occupancy of each face plus a threading path	How packaging signals change the answer

This document is not a summary of the four existing Module 4 documents. It is the derivation chain, written once, with the layers' points of contact and their one published disagreement made explicit. **§9 is the wet-lab contract**: which measurement enters, which parameter it moves, and what the model is obliged to do when the number disagrees.

Each layer, the bridge between them and the fitting routine ship as **runnable notebooks** (`M4_L1_rate_equations.ipynb`, `M4_L2_nucleation_phase_diagram.ipynb`, `M4_L3_coarsegrained_MC.ipynb`, `M4_L4_packaging_signal_coassembly.ipynb`, `M4_bridge_CG_to_ODE_to_bench.ipynb`, `M4_FIT_to_bench_data.ipynb`). Each writes out the module it needs and runs top to bottom on a fresh machine; see their `README.md`. The experimental designs, literature landscape and Spanish-language team material live in their own artifacts and are referenced, not repeated.

Every equation below is either derived here from a stated physical assumption or lifted verbatim from a cited paper. Where a number appears, it was read from the code that produces it, not from memory — the project rule is that no value reaches a deliverable without resolving against its source.

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1. Notation and the constants shared by all four layers

Throughout, a **subunit** means one coat-protein dimer, the obligate structural unit of MS2 established crystallographically by Golmohammadi and colleagues and refined for recombinant assembly by Ni and colleagues. A T=3 shell contains

$$N = 90 \text{ dimers} = 180 \text{ coat} - \text{protein monomers}$$

Free energies are in kcal/mol and the thermal scale is

$$RT = 0.593 \text{ kcal/mol} \quad (T = 310 \text{ K})$$

so 1 kcal/mol is 1.69 k_BT. Concentrations are in μM unless stated. The sign convention is thermodynamic: $\Delta G < 0$ favours the bound state, and a dissociation constant maps to a binding free energy by

$$\Delta G_{bind} = RT \ln K_d$$

with K_d expressed relative to the standard state. This single relation is what lets a measured affinity enter a kinetic model, and it is used in L1, L4 and Module 5 alike.

Two energies recur and must be kept distinct:

- $g_{assoc} > 0$ — the **association penalty** per subunit added: the translational and rotational entropy a free dimer surrenders on joining a shell. Zlotnick's foundational analysis of icosahedral assembly identifies this term as the origin of the nucleation barrier.
- $g_{contact} < 0$ — the free energy of **one subunit-subunit contact**. Ceres and Zlotnick measured it directly for hepatitis B capsids and found it remarkably weak, a few kcal/mol, which is precisely why assembly is reversible enough to be productive.

Assembly is possible because a subunit joining an existing shell forms *more than one* contact, so a single penalty is paid against several favourable contacts. Everything in Module 4 is an elaboration of that one sentence.

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2. Layer L1 — the nucleation-elongation cascade

2.1 Upstream: the monomer-dimer equilibrium

Module 4 takes a total coat-protein concentration from Module 3 and must convert it to *free dimer*, since the dimer is the assembling species. With dimerization

$$2\,M \rightleftharpoons D, \qquad K_d^{dim} = \frac{[M]^2}{[D]}$$

and total protein $P_{tot} = [M] + 2[D]$, eliminating $[D]$ gives a quadratic in $[M]$ whose physical root is

$$[M] = \frac{-K_d^{dim} + \sqrt{(K_d^{dim})^2 + 8K_d^{dim}P_{tot}}}{4}$$

MS2 dimerization is tight — the folded dimer is the thermodynamic ground state of the isolated chain — so with $K_d^{dim} = 0.01$ μM and $P_{tot} = 8$ μM the partition is $[D] = 3.90$ μM, **97.5% of protein mass already dimeric**. Loosening K_d^{dim} by two orders of magnitude to 1.0 μM only drops that to 77.9%. The conclusion transfers to the model as a licence: within the plausible range for MS2, Module 4 may take $[D]_0 \approx P_{tot}/2$ and ignore the

monomer pool. Module 3's steady state ($m_{ss} = 38.5$ nM mRNA, 7.96 μM monomer-equivalent) therefore enters Module 4 as $[D]_0 = 3.98$ μM.

2.2 The state space and the master equation

Let $x_j(t)$ be the molar concentration of shells containing exactly j subunits, with x_1 the free dimer. Growth proceeds one subunit at a time — the assumption that no two partial shells fuse, justified for capsid assembly by the strong asymmetry between the abundant free subunit and the rare intermediate. Then the flux through step $j \to j+1$ is

$$J_j = f_j x_j x_1 - b_{j+1} x_{j+1}$$

and the cascade is the N-dimensional system

$$\frac{dx_j}{dt} = J_{j-1} - J_j$$

closed by $J_0 \equiv 0$, $J_N \equiv 0$ (a complete shell is inert), and by the free-subunit balance which must count every subunit consumed anywhere in the ladder:

$$\frac{dx_1}{dt} = -\sum_{j=1}^{N-1} J_j - J_1$$

Total mass $\sum_j j\,x_j$ is conserved by construction; in the production runs it holds to 1.5×10^{-15} relative error, which is the numerical check that the flux bookkeeping is right.

2.3 The contact ladder — where the geometry enters

The forward rate is a diffusion-limited on-rate, taken size-independent:

$$f_j = k_a \quad \text{for all } j$$

All the structure lives in the reverse rates, and they are **not free parameters**. A subunit added to make the j-mer forms $c(j)$ new contacts, and the free-energy change of that addition is

$$\Delta G_{add}(j) = g_{assoc} + c(j)\, g_{contact}$$

The production model encodes $c(j)$ as a three-regime step function reflecting the lattice: an isolated subunit landing on a small cluster has one neighbour; a subunit filling the edge of a growing cap has two; a subunit closing one of the final vertices is surrounded on three sides.

$$c(j) = 1 \;\; (j \le n_c), \quad c(j) = 2 \;\; (n_c < j < N - 8), \quad c(j) = 3 \;\; (j \ge N - 8)$$

With the literature energies $g_{assoc} = +6.0$ and $g_{contact} = -4.0$ kcal/mol this gives a ladder that is **uphill before the nucleus and downhill after it**:

Regime	Contacts c(j)	ΔG(add)	Meaning
$j \le 4$	1	**+2.0** kcal/mol	Nucleation is rare — this is the barrier
$4 < j < 82$	2	**–2.0** kcal/mol	Elongation is favourable and fast
$j \ge 82$	3	**–6.0** kcal/mol	Closure is nearly irreversible

2.4 Detailed balance fixes the reverse rates

At equilibrium each step must satisfy $f_j x_j x_1 = b_{j+1} x_{j+1}$, so the ratio of rates is the equilibrium constant of that step:

$$\frac{b_{j+1}}{f_j} = K_d(j) = \exp\left(\frac{\Delta G_{add}(j)}{RT}\right)$$

hence

$$b_{j+1} = k_a \exp\left(\frac{\Delta G_{add}(j)}{RT}\right)$$

This is the defensible core of Module 4. Ninety reverse rates are determined by two energies and a contact count. Nothing in the ladder can be tuned to produce a desired yield curve without moving g_{assoc} or $g_{contact}$ out of their independently measured ranges. A model that fitted 90 off-rates could reproduce any observation and would predict nothing.

The cumulative profile follows by summation. With $B(j) = \sum_{i<j} c(i)$ the total bonds in a j-mer,

$$\Delta G(j) = j\, g_{assoc} + B(j)\, g_{contact}$$

whose maximum defines the critical nucleus. For the step ladder that maximum sits at $j = n_c = 4$ by construction — an *input*. Section 5 replaces it with an emergent value.

2.5 What L1 predicts: the concentration optimum and the trap

Integrating the cascade to $t = 240$ min at 37 °C over initial dimer concentration:

$[D]_0$ (μM)	Closed capsid	Free dimer	Partial shells	Mean partial size
0.50	7.9%	88.1%	4.0%	20
1.00	41.9%	54.4%	3.7%	27
2.00	77.4%	21.6%	1.0%	20
2.48	**94.8%**	2.4%	2.8%	76
3.00	52.5%	1.1%	46.4%	65
5.00	0.1%	0.7%	99.2%	48
10.00	0.0%	0.3%	99.7%	29

More dimer is not better. The optimum is interior, at $[D]_0 = 2.48$ μM. The mechanism of the collapse is visible in the columns: between 2.5 and 5 μM the yield falls from 95% to nil *while free dimer stays below 1%*. The protein has not failed to react — it has all reacted, into partial shells that can no longer find the subunits they need. That is **kinetic trapping**, and it is the physically interesting content of the layer. Zlotnick and Endres established the mechanism; Hagan and Chandler's simulations connect it to the same rate-equation structure used here.

The existence of an interior optimum is also the model's principal falsifiable claim. A monotonically saturating yield curve would break the detailed-balance construction and both scales built on it.

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3. Layer L2 — classical nucleation theory as the widening layer

L1 answers "how much, when" for one parameter set. L2 answers a different question — **for which contact energies and concentrations is productive assembly possible at all** — by coarse-graining the ladder into a continuous droplet free energy. The gain is a phase diagram; the cost is that discreteness and geometry are lost.

3.1 The droplet free energy

Treat cluster size n as continuous. Growth is driven by a per-subunit chemical potential difference and opposed by the unsatisfied bonds at the cluster rim. Bulk scales as n, rim as the perimeter, which for a two-dimensional cap on a shell scales as $n^{2/3}$ in the CNT convention:

$$G(n) = -n\,\Delta\mu + \sigma a_1 n^{2/3}$$

with σ an edge tension and a_1 a geometric prefactor. The drive combines the contact energy with the entropic cost of removing a subunit from solution:

$$\Delta\mu = -g_b\,\bar{c} - RT\ln\left(\frac{c_0}{c}\right)$$

where $g_b < 0$ is the energy per contact, $\bar{c}$ the net contacts a subunit gains on joining, and c_0 a reference concentration. The logarithm is the supersaturation: **below a threshold concentration $\Delta\mu \le 0$ and no cluster is stable at any size.**

3.2 Barrier and critical nucleus

$G(n)$ rises, peaks, and falls. Setting $dG/dn = 0$:

$$-\Delta\mu + \tfrac{2}{3}\sigma a_1 n^{-1/3} = 0 \Longrightarrow n^* = \left(\frac{2\sigma a_1}{3\Delta\mu}\right)^3$$

and substituting back,

$$G^* = \frac{4}{27}\,\frac{(\sigma a_1)^3}{\Delta\mu^2}$$

Both scalings are informative in themselves: the critical nucleus shrinks as the inverse cube of the drive and the barrier as its inverse square. **Doubling supersaturation cuts the barrier fourfold and the nucleus eightfold** — which is why capsid yield is so violently concentration-sensitive, and why the optimum in L1 is sharp rather than broad.

3.3 Three regimes, and a phase diagram

Classifying each point of the (g_b, C) plane by the barrier it produces:

Regime	Condition
no assembly	$\Delta\mu \le 0$, or $G^* > 35$ kcal/mol
kinetic trap	$G^* < 1$ kcal/mol
productive	otherwise

The two thresholds are timescale statements, not new physics: a barrier above ~ 35 kcal/mol is not crossed in an incubation, and a barrier below ~ 1 kcal/mol is no barrier at all, so nucleation outruns elongation and every subunit starts a shell that none can finish. Over $g_b \in [-1, -5]$ kcal/mol per contact and $C \in [0.3, 300]$ μM, with $\bar{c} = 1.0$, $\sigma = 1.6$ kcal/mol, $a_1 = 1.5$, $C_0 = 500$ μM:

- **no assembly 40%** of the plane,
- **productive 27%**,
- **kinetic trap 33%**.

A representative productive point, $g_b = -1.5$ kcal/mol and $C = 120$ μM, gives $\Delta\mu = 0.65$ kcal/mol, $n^* = 14.7$ dimers and $G^* = 4.79$ kcal/mol $= 8.1\,k_BT$ — a barrier crossed often enough to nucleate, rarely enough not to exhaust the pool.

The headline is that productive assembly occupies only about a quarter of the accessible parameter plane, and is bracketed on both sides by failure. Assembly conditions are not a threshold to exceed but a window to hit. Note that L2's $n^* \approx 15$ is much larger than L1's assumed $n_c = 4$; the two are not directly comparable, because L2's continuum rim term overestimates the penalty for the very small clusters where discreteness dominates. Reconciling them is exactly what L3 is for.

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4. Layer L3 — coarse-grained Monte Carlo: measuring what L1 assumes

L1 and L2 both take the geometry as given: a contact count, an edge tension. L3 puts 90 subunits in a box with explicit positions, orientations and a directional interaction, and lets the geometry emerge.

4.1 A patchy-particle potential with built-in curvature

Each subunit is a hard sphere of diameter σ decorated with attractive patches in the Kern-Frenkel form: the interaction is switched on only when both partners point at each other within an angular tolerance δ. For a pair at separation r with unit vector $\hat{d}$,

Configuration of the pair	Pair energy u(i,j)
$r < \sigma$ (hard core)	$+\infty$
$r > r_{well}$ (out of range)	0
$\sigma \le r \le r_{well}$, patches aligned	$-\varepsilon\left(\frac{1}{2} + \frac{1}{2}\chi_{ij}\right)$
$\sigma \le r \le r_{well}$, patches not aligned	0

"Patches aligned" means, with $\hat{p}$ the patch vectors in the laboratory frame,

$$\max_k \hat{p}_i^{(k)} \cdot \hat{d} > \cos\delta \quad \text{and} \quad \max_l \hat{p}_j^{(l)} \cdot (-\hat{d}) > \cos\delta$$

with $\delta = 28°$ and $r_{well} = 1.2\,\sigma$.

The factor χ_{ij} is what makes this a *capsid* model rather than a generic self-assembling fluid. A flat sheet is the ground state of a naive patchy sphere; a closed shell requires a preferred dihedral angle between neighbours. Rewarding anti-alignment of the engaged patch vectors,

$$\chi_{ij} = \frac{1}{2}\left(1 + \hat{p}_i^{(k)} \cdot \left(-\hat{p}_j^{(l)}\right)\right)$$

builds the spontaneous curvature in energetically, so growing caps close instead of extending. The patch geometry itself is laid out as a curved five-patch wedge, the local environment of a T=3 lattice site.

4.2 Sampling

Configurations are sampled by Metropolis Monte Carlo in the canonical ensemble, with translation and rotation moves accepted at

$$P_{acc} = \min\left[1, \exp\left(-\frac{\Delta U}{k_B T}\right)\right]$$

and a cell list restricting neighbour search to $r < r_{well}$, which is what makes 90 particles over 1200 sweeps tractable. Clusters are defined by a bond threshold at $-0.4\,\varepsilon$, i.e. a pair counts as bonded only if it is well inside the attractive well rather than grazing it.

4.3 What L3 measures

Scanning volume fraction ϕ at $\varepsilon = 11 \quad k_B T$:

Volume fraction ϕ	Largest cluster (subunits)	Mean coordination	Well-formed fraction
0.03	12.7	1.75	0.226
0.06	23.0	1.89	0.244
0.10	35.7	2.13	0.393
0.15	58.3	2.07	0.393
0.22	47.0	1.75	0.304
0.30	38.3	1.38	0.211

The optimum in ϕ is interior — near 0.10–0.15 — and it is not the same statement as L1's optimum in concentration, but it is the same physics. Crowding first accelerates nucleation, then produces more nuclei than there are subunits to complete them, and mean coordination *falls* as the largest cluster shrinks: the signature of malformed aggregates rather than shells.

L3 is where the assumption of L1 becomes a measurement, which is the subject of the next section.

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5. The bridge L3 → L1, and the disagreement we chose to publish

5.1 A one-parameter cap law

The bonds in a compact cap of j subunits on a closed shell of N can be written as a bulk term minus the unsatisfied open perimeter. In a T=3 shell each dimer has five neighbours, so a closed shell holds $\frac{5}{2}(N-1)$ bonds per the handshake count; the rim correction must vanish at both $j=1$ and $j=N$, which the geometric mean of occupied and vacant fractions does. Hence

$$b(j) = \tfrac{5}{2}(j-1) - a\left[\sqrt{j\left(1-\tfrac{j}{N}\right)} - s_1\right]$$

with s_1 chosen so $b(1)=0$ exactly and a **single** shape parameter a carrying the rim penalty. Fitting a to 2,547 clusters censused from the L3 trajectories at $\varepsilon = 6, 7, 8$:

$$a = 6.75 \pm 0.37 \quad \text{(RMS residual 4.8 bonds)}$$

and the law extrapolates to $b(90) = 229$ against the geometric 225 — a 1.8% closure error from a fit that never saw a complete shell. The differentiated ladder

$$c(j) = b(j+1) - b(j)$$

is the measured analogue of L1's step function. **It is not a step function.** It rises continuously, from $c \approx 0.7$ at $j=4$, through $c \approx 1.5$ at $j=10$, to the bulk value at large j.

5.2 An emergent critical nucleus

With a measured ladder, n_c stops being an input. It is the argmax of the cumulative profile:

$$n_c = \arg\max_j \left[j\, g_{assoc} + b(j)\, g_{contact} \right]$$

Substituting the measured a at *unchanged* literature energies abolishes assembly outright: the barrier becomes 24 kcal/mol $\approx 41\, k_B T$ and nothing nucleates in 240 minutes. That is not a bug but the layer's most instructive output — **the ladder shape and the energy offsets are not independently transferable**, because g_{assoc} was originally calibrated against the step ladder.

Enforcing equal barrier height instead requires $g_{assoc} = 4.01$ kcal/mol — and at that value the CG geometry independently returns

$$n_c = 4$$

the very value L1 assumed. A genuine cross-scale consistency check, on a quantity neither layer was tuned to reproduce, and it passes. It also agrees with the 3–5 dimer nucleus inferred from single-particle waiting times by Garmann and colleagues.

5.3 The disagreement

Matching barrier *height* is not matching ladder *shape*, and the shape is what sets the concentration optimum:

Model	Predicted optimal $[D]_0$	Yield at optimum
Step ladder (production L1)	2.50 μM	95%
CG-measured ladder, $g_{assoc} \approx 4.0$	9.25 μM	97%

The two scales disagree by 3.7× on the single number the wet lab will set. Across the literature box for the energies the CG-bridged optimum spans 7.8–15 μM and never falls below 5.8 μM. Inverting the other way — demanding the step-ladder optimum of 2.5 μM from the measured geometry — requires $g_{contact} = -5.55$ kcal/mol, outside Ceres and Zlotnick's independently measured -4.4 to -2.9 interval.

Three sources of evidence cannot all be right. We publish the disagreement rather than picking the flattering branch, because one benchtop concentration sweep adjudicates it: a peak near 8–12 μM vindicates the CG geometry, near 2.5 μM indicts it, and no peak at all falsifies the detailed-balance construction underneath both.

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6. Layer L4 — packaging-signal-mediated co-assembly

L1–L3 model protein alone. The BruSafe particle assembles *around an mRNA carrying engineered pac-sites*, and Dykeman, Stockley and Twarock showed that such signals change assembly qualitatively, not just quantitatively. L4 is an independent re-implementation of their framework on a 12-face study system, validated on exact invariants before any kinetics were compared.

6.1 Two reaction orders, both at detailed balance

Two classes of event occur. A subunit can bind a packaging signal on the cargo — second order, since it consumes free protein — or two adjacent bound subunits can form a protein-protein bond — first order, an intramolecular closure. Detailed balance is imposed separately on each. For a signal of affinity ΔG_R at cargo concentration C ,

$$k_{2R} = \frac{k_{1R}}{C \exp\left(-\Delta G_R / RT\right)}$$

and for a protein-protein bond of energy ΔG_P ,

$$k_{2P} = k_{1P} \exp\left(\frac{\Delta G_P}{RT}\right)$$

Because subunit count scales with ensemble size at fixed volume, the on-rate must be rescaled against the reference ensemble to hold concentration constant:

$$k_{1R} \to k_{1R} \frac{N_{RNA}}{N_{ref}}, \quad N_{ref} = 3000$$

Events are drawn by the Gillespie algorithm — with total propensity A , the waiting time is

$$\tau = -\frac{\ln u}{A}, \quad u \sim \mathcal{U}(0, 1)$$

and propensities are held in a binary-indexed tree so that selecting an event costs $O(\log N)$ rather than $O(N)$.

6.2 The trap is geometric, not energetic

The mechanism that makes this layer non-trivial: the cargo **threads the shell as a self-avoiding walk on the face-adjacency graph**. A growing end can reach a face all of whose neighbours are already visited — a dead end. The shell is then incomplete and cannot be finished without reversing, and reversal means paying k_{2P} to break protein bonds. The trap therefore arises from the topology of the threading path, not from an energy barrier. Two earlier versions of this engine were discarded for getting exactly this wrong: one allowed free rerouting and produced near-complete yield in every condition (nothing ever trapped), the other gated escape on an invented reserved-face rule and over-trapped.

6.3 Validation on exact invariants

Before comparing any kinetics, two combinatorial quantities of the published model were reproduced exactly:

- **73** unique protein-only intermediates,
- **1,264** distinct threading paths, from 75,840 directed Hamiltonian paths divided by the order-60 icosahedral rotation group.

Two exact agreements on quantities that depend only on the geometry and the assembly rules are the licence to compare rates at all.

6.4 What L4 predicts

Finding	Numbers
Interior optimum in signal affinity, 12-PS cargo	96.5% at $\Delta G_{PS} = -2.0$, falling to 54.3% at -12
Optimum reverses for a sparse 3-pac-site cargo	44.5% at -2.0, rising to **60.1% at** -9.0
Signal count is the largest unused lever	20.3 / 44.5 / 60.8 / 96.5% at 1 / 3 / 6 / 12 sites (≈14 points per site at $n = 3$)
Layout matters at matched mean affinity	300 random cargoes span 63.3–97.5% (mean 80.2, sd 6.1)
Protein-protein window is narrow	95.1% at $\Delta G_P = -2.0$; 42.8% at -3.0; 18.3% at -4.0

The reversal is the result that matters for BruSafe and is **not present in the source paper**, which only ever simulated fully decorated genomes. For a densely decorated cargo, very tight signals are harmful: they nucleate everywhere and freeze partial shells. For a sparse cargo — three pac-sites, as in our cassette — there is no competing nucleation to suppress, and tighter is better. The MS2 TR hairpin itself sits at $\Delta G = RT \ln K_d = -11.6 *to* -12.3 *kcal/molfor* K_d = 1$–3 nM, i.e. at the tight end. **The theory therefore supports keeping the wild-type recognition core rather than weakening it**, which is what the de-repeated triple-pac-site design does.

What L4 does *not* reproduce is stated for the record: the paper's absolute yields (their dimensionless conversion factor is described only as adjusted, never given, so absolute yields are underdetermined by the published text), their -3.0 kcal/mol optimum, and their nucleator-position preference. These are flagged, not tuned.

7. The parameter registry

Every quantity Module 4 consumes, its value, its literature bracket, and the measurement that would pin it down. No value in this table is a free fit.

Parameter	Value	Literature range	Identified by	Source
$g_{contact}$	−4.0 kcal/mol	−4.4 to −2.9	Position of the yield optimum	Ceres & Zlotnick 2002
g_{assoc}	+6.0 kcal/mol	4 to 8	Held fixed; absorbed into $g_{contact}$ if floated	Zlotnick 1994
k_a	3.0 $\mu M^{-1}min^{-1}$	0.4 to 4	Peak height; nucleation lag	Hagan & Chandler 2010
n_c	4 dimers	3 to 6	Single-particle waiting times	Garmann 2022
N	90 dimers	structural, not fitted	Crystallography	Zlotnick 1994; Ni 1995
K_d^{dim}	0.01 μM	0.001 to 1	Analytical ultracentrifugation	Ni 1995
σ (L2 edge tension)	1.6 kcal/mol	calibrated	Barrier height	derived, this work
$\bar{c}$ (L2 net contacts)	1.0	calibrated	Drive slope	derived, this work
a (cap-law shape)	6.75 ± 0.37	measured	CG bond census, 2547 clusters	derived, this work
ΔG_P (L4)	−2.5 kcal/mol	−2.0 to −4.0	Yield vs protein-protein energy	Dykeman 2013
ΔG_{PS} (L4)	−2.0 to −12 kcal/mol	measured for TR	Filter binding, ITC	Dykeman 2013; Stockley 2016
RT	0.593 kcal/mol	310 K	—	—

8. Limits of validity

Stated plainly, because a model whose limits are hidden cannot be trusted where it does work.

1. **L1 is a mean-field rate equation.** It tracks concentrations of size classes, not individual shells, so it cannot distinguish a well-formed 90-mer from a malformed one of the same size. Its yield is properly read as an **upper bound on well-formed particles**. Hagan's review of assembly modelling makes this limitation explicit for exactly this class of model.
2. **One-subunit-at-a-time growth** excludes fusion of partial shells. Justified while intermediates are rare; it is least safe deep in the trapped regime, which is where the model's most dramatic prediction lives.
3. **L2's continuum rim term** is wrong at small n, where discreteness dominates. Its $n^* \approx 15$ should be read as a scaling statement, not a prediction of the nucleus.
4. **L3 is a 90-particle model in reduced units**, with a patch geometry that approximates a T=3 lattice site rather than reproducing the MS2 interface. Its ϕ optimum transfers as a direction, not a molar concentration.
5. **L4 runs on a 12-face study system**, not the T=3 lattice. Its ordering and reversals transfer; its magnitudes do not. Porting it to 90 faces is the natural extension.
6. **No electrostatics.** Cargo charge, ionic strength and pH enter only implicitly through effective energies, whereas Chevreuil and colleagues show the assembly pathway itself can depend on them.
7. **Temperature enters only through RT.** The hydrophobic, entropy-driven character of the interface means the energies themselves are temperature-dependent; the temperature lever is capped accordingly.
8. **The two false retraction flags** in the corpus (`10.1371/journal.pcbi.1010652`, `10.1371/journal.pcbi.1009913`) are `new_version` updates re-verified against Crossref, not retractions, and must be cleared before the corpus is cited as evidence of screening.

9. Wet-lab validation: where this model meets the bench

Everything above is Mode A — prediction with no data in it. This section is the contract between the model and the wet lab: which measurements enter, which parameter each one moves, and what the model is obliged to do when the number comes back wrong. It is written for the equipment actually available: a plate reader, a gel rig, a spectrophotometer, and **at most three TEM sessions**, which is the binding constraint on the whole design.

9.1 The one experiment that matters most

A concentration sweep of coat-protein dimer, read by turbidity, with TEM at three points.

Every layer of Module 4 makes the same qualitative claim — that yield is **non-monotonic** in dimer concentration, rising to an optimum and then collapsing into kinetic traps. L1 derives it from rate equations, L2 explains it as a moving barrier, L3 reproduces it from geometry alone. A single sweep tests all three at once. If yield instead saturates monotonically, the central claim of the module is wrong, and no amount of re-fitting rescues it.

Design: assemble at $[D]_0 \in \{0.5, 1, 2, 3, 5\}$ μM (add 8 and 12 μM if protein allows), 240 min, three replicates, one buffer. Read A_{340} continuously in the plate reader.

9.2 Why turbidity alone is not enough

Turbidity is a mass-squared signal: for particles small against the wavelength, scattering per species goes as (mass) ^{2} , so the plate reader reports

$$\tau(t) \propto \sum_j j^2 x_j(t),$$

which a **partial shell answers almost as loudly as a closed one**. A 60-dimer malformed aggregate scatters $(60/90)^2 = 44\%$ as much as a complete capsid. This is precisely the kinetic-trap signature: above the optimum the model predicts turbidity that *plateaus high* while the closed-capsid count *collapses to zero*. A plate reader on its own cannot tell that apart from success, which is the entire reason TEM sessions must be spent, and spent in the right places.

Native PAGE is the cheap partial fix: closed capsids run as a single sharp species, trapped intermediates as a smear. It resolves the ambiguity qualitatively for every concentration in the sweep, at no TEM cost.

9.3 Spending three TEM sessions

The three sessions are the scarce resource, so they are allocated to the three points where the model's predictions differ most, not spread evenly across the sweep:

Session	$[D]_0$	Model predicts	What it discriminates
1	at the **turbidity peak**	mostly closed 27 nm shells	confirms the optimum is productive, not just scattering
2	**2× above the peak**	high turbidity, few closed shells, mass in large partial shells	the trap prediction — the single most falsifiable claim in the module
3	**~0.4× the peak**	few particles, most mass free dimer	confirms the low side fails by *not nucleating*, not by trapping

Session 2 is the one that earns the module its keep. Do not sacrifice it to a replicate.

Count ≥ 200 particles per grid and score each as correct-diameter (25–29 nm) or not; the fraction is what the model consumes. Record the count itself — it sets the binomial error bar, and a fraction with no n behind it cannot be weighted in a fit.

9.4 What to graph, and what each graph is for

1. **Turbidity traces, A_{340} vs time, one curve per concentration.** Read the *lag* and the *plateau*, not just the endpoint. Lag length is a nucleation observable; plateau height is a mass observable. Generated by the bridge notebook as `M4_bridge_observables.png`.
2. **Yield curve: capsid fraction vs $[D]_0$, log-x.** The headline figure. The model's optimum position and height are read straight off it. `M4_L1_yield_curve.png`.
3. **TEM size histogram at the three sessions, mass-weighted, log-y.** The trap is visible as a mid-size hump that has no business being there. A linear y-axis hides it.
4. **Prior versus posterior yield curve with your measurement on it.** This is the figure for the wiki: it shows the model *moving* because of your data. `M4_bridge_correction.png` and `M4_FIT_prior_vs_posterior.png`.
5. **Profile-likelihood curve for the contact energy.** Turns a fitted number into an interval, and shows a judge that the fit was interrogated rather than declared. `M4_FIT_profile.png`.

9.5 How the model is adjusted afterwards — which observable moves which parameter

The fit is identifiable only because different observables carry different parameters. This was checked before any data existed, by scanning each parameter across its literature range and recording what moved:

Observable	Parameter it constrains	Evidence it is the right pairing
Position of the yield optimum	g_{contact}	across its literature range the optimum moves 12.0 → 0.44 μM
Height of the optimum, and lag length	k_a	across its range the position stays at 2.52 μM while height moves 0.39 → 0.95
Overall vertical offset	s_{rec}, a recovery scale	absorbs purification and grid-sampling losses that are not physics
TEM size distribution above the optimum	validates n_c and the ladder	not fitted — a genuine out-of-sample test

Because position and height are carried by different parameters, two numbers — the optimum's location `D_MEAS` and its height `Y_MEAS` — invert to two energies. That inversion is the corrective step in `M4_bridge_CG_to_ODE_to_bench.ipynb`; the fuller weighted fit over the whole curve, with the profile interval, is `M4_FIT_to_bench_data.ipynb`.

n_c is deliberately held fixed at 4 in the fit. From concentration data alone it is confounded with g_{contact} — a larger nucleus and a weaker contact shift the yield curve the same way — so floating both would produce a tighter-looking fit that constrains neither. The value 4 is the literature prior, not a measurement.

The bridge measures n_c independently, from where the coarse-grained contact ladder turns the free-energy barrier over, and reads $n_c = 8$. **That factor of two is a real disagreement between the two routes, and we report it as one.** It is not reconciled by tuning either number: the prior is a mean-field estimate for a $T = 3$ shell, the bridge value comes from one patch geometry at one bond energy, and neither has yet been checked against bench data. What the disagreement does mean is that any yield-curve fit quoting a contact energy must also quote the n_c it assumed — which is why the profile interval in `M4_FIT_to_bench_data.ipynb` is reported at fixed n_c rather than marginalised over it.

9.6 What we do if the data disagrees

Stated in advance, so the response is not chosen after seeing the result:

- **Optimum in the wrong place** → re-fit g_{contact}. Expected; this is what the fitting notebook is for.
- **No optimum at all, monotonic saturation** → L1's mechanism is wrong for MS2. Report it. The nucleation-elongation picture would not survive that, and dressing it up would be worse than losing it.
- **Optimum present but TEM above it shows free dimer rather than partial shells** → the yield collapse is real but the *mechanism* is not trapping. L1 keeps its curve and loses its explanation.
- **Fitted g_{contact} outside the –2.9 to –4.4 kcal/mol literature interval** → this has already happened. The bridge inverts to **–4.78 kcal/mol**, and the notebook prints the comparison as a plain `False`. It is reported as a disagreement (§5.3), not tuned away.

9.7 Data format

`M4_bench_data_template.csv` carries the column names both data-facing notebooks read: `assay, D0_uM, t_min, replicate, value, value_sd, units, n_particles, notes`, with `assay` one of `turbidity`, `nativePAGE`, `TEM`. Fill it in, point the fitting notebook at it, and every figure in §9.4 regenerates. `M4_bench_data_example.csv` is synthetic and exists only so the notebooks run before data exists; it must be replaced, not fitted.

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10. What would falsify each layer

Layer	Prediction	Falsified by
L1	Interior optimum in $[D]_0$, with collapse at high concentration and free dimer staying low	A monotonically saturating yield curve
L1	Trapped mass sits in large partial shells (78–79 dimers at 2.5 μM, 69–71 at 3.0, 48–50 at 5.0)	TEM showing free dimer, not partial shells, above the optimum
L2	Productive assembly occupies a bounded window, not a threshold	Productive yield at arbitrarily high concentration
L3 → L1	Optimum at 8–12 μM	Optimum at 2.5 μM (indicts CG geometry) or no peak (breaks both)
L4	Tighter pac-sites help a sparse cargo	Yield falling monotonically with pac-site affinity at $n_{PS} = 3$

The concentration sweep is the single highest-information experiment in Module 4: turbidity across 1–20 μM dimer, with TEM at three points, because a turbidity plateau *without* a matching capsid count is the trap signature and turbidity alone cannot see it. The full design is §9.

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11. Bibliography

All entries below are Crossref-verified; the full 70-record corpus with themes and verification columns is `M4_corpus_verified.csv`.

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